

Enterprise Campus Network Design & Implementation Report

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1.Executive Summary

Enterprise 3-tier Network Infrastructure

Designed and deployed a resilient, scalable campus network for a mid-sized, multi-building enterprise using a traditional Core–Distribution–Access architecture. The design delivers a structured, highly available foundation engineered to minimize single points of failure, provide controlled edge connectivity, and support future organizational growth.

Key Technical Highlights

- **High Availability:** LACP EtherChannel for bandwidth aggregation and link redundancy, with HSRP providing dual-gateway failover.
- **Dynamic Routing & Segmentation:** Single-area OSPF for dynamic routing and path convergence, alongside strict multi-VLAN segmentation and isolation.
- **Edge Connectivity:** PAT/NAT Overload providing scalable Internet connectivity for private internal hosts.

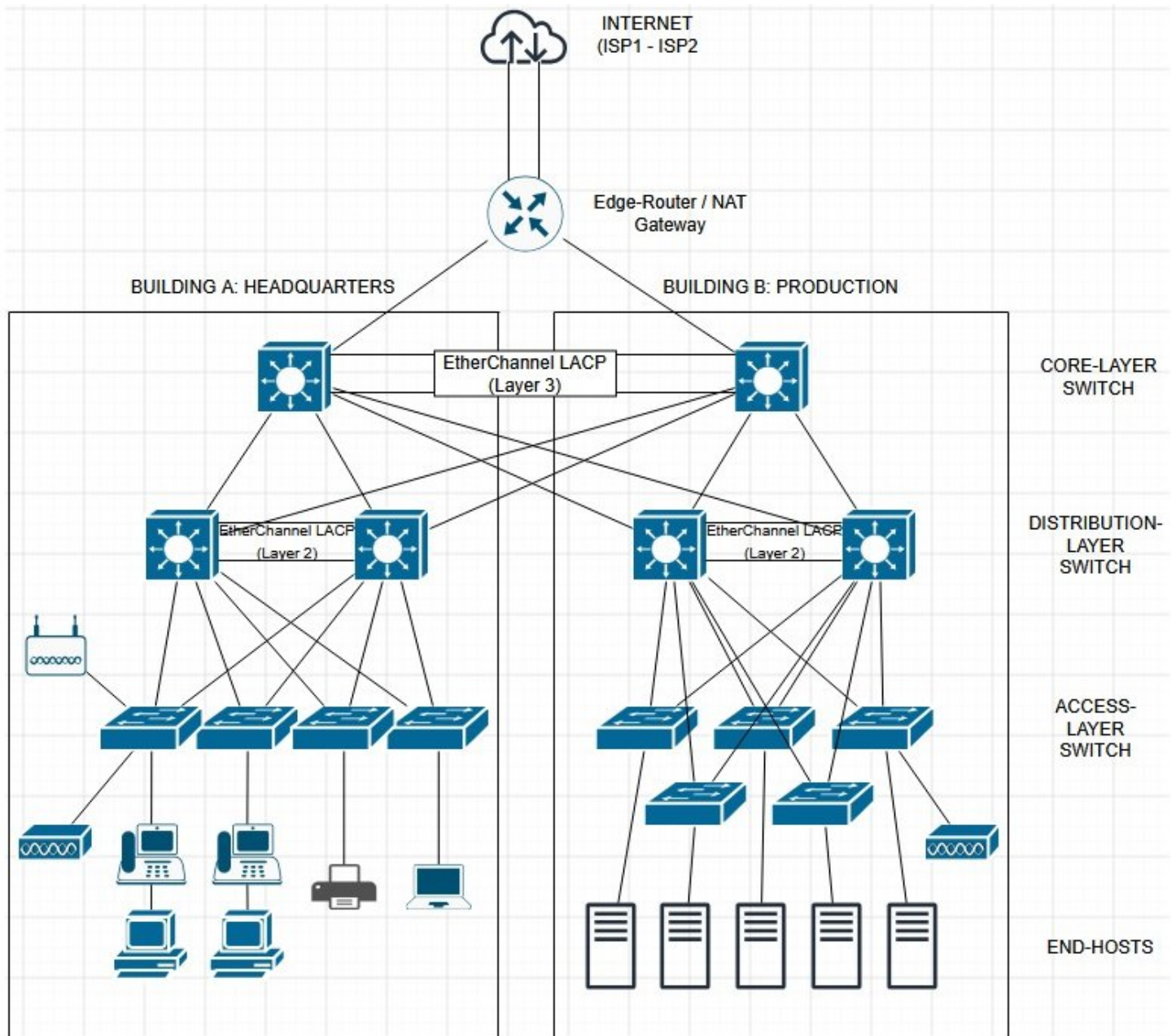
2.Business Requirements

To support the operational demands of a growing, multi-building enterprise, the network infrastructure was designed around core business imperatives: maximizing application availability, prioritizing real-time voice communications, enforcing strict data compliance, and isolating untrusted guest access.

Requirement	Business Reason / Driver	Technical Implementation
High Availability (HA)	Ensures business-critical applications remain online during link or hardware failures.	Deployed HSRP for dual-gateway failover and LACP EtherChannel for physical link redundancy.
Voice VLANs	Guarantees Quality of Service (QoS) and low latency for the enterprise IP telephony rollout.	Configured dedicated Voice VLANs to segregate VoIP traffic from standard data streams.
Network Segmentation	Enforces security compliance by restricting inter-departmental traffic.	Implemented multi-VLAN segmentation paired with Access Control Lists (ACLs).
Guest Wi-Fi	Provides secure internet access for visitors without exposing internal enterprise resources.	Isolated guest traffic onto a dedicated VLAN with direct, restricted NAT transit to the edge.

3. Network Topology Diagrams

3.1 High-Level Topology

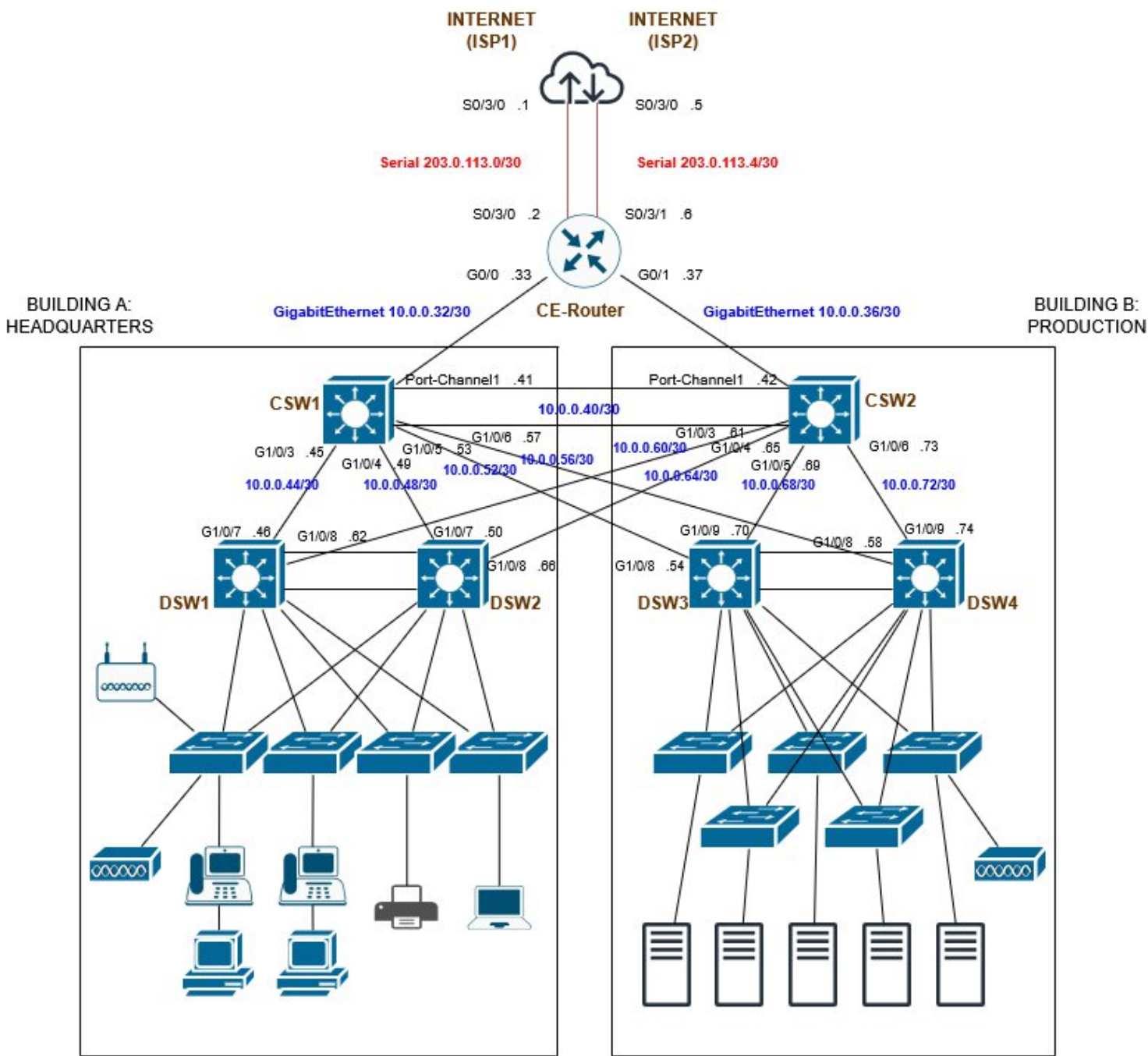


3.2 Physical Topology

R1	G0/0	CSW1 - G1/0/7	DSW1	G1/0/3	ASW1 - G0/1	ASW1	F0/1	PC1 & Phone1
	G0/1	CSW2 - G1/0/7		G1/0/4	ASW2 - G0/1		F0/2	LWAP1
	S0/3/0	ISP-1 - S0/3/0		G1/0/5	ASW3 - G0/1		F0/3	WLC1
	S0/3/1	ISP-2 - S0/3/0		G1/0/6	ASW4 - G0/1		G0/1	DSW1 - G1/0/3
			PortChannel1	G1/0/1	DSW2 - G1/0/1		G0/2	DSW2 - G1/0/3
				G1/0/2	DSW2 - G1/0/2	ASW2	F0/1	PC2 & Phone2
CSW1	G1/0/7	R1 - G0/0		G1/0/7	CSW1 - G1/0/3		G0/1	DSW1 - G1/0/4
PortChannel1	G1/0/1	CSW2 - G1/0/1		G1/0/8	CSW2 - G1/0/3		G0/2	DSW2 - G1/0/4
	G1/0/2	CSW2 - G1/0/2	DSW2	G1/0/3	ASW1 - G0/2	ASW3	F0/1	Printer1
	G1/0/3	DSW1 - G1/0/7		G1/0/4	ASW2 - G0/2		G0/1	DSW1 - G1/0/5
	G1/0/4	DSW2 - G1/0/7		G1/0/5	ASW3 - G0/2		G0/2	DSW2 - G1/0/5
	G1/0/5	DSW3 - G1/0/8		G1/0/6	ASW4 - G0/2	ASW4	F0/1	HR
	G1/0/6	DSW4 - G1/0/8	PortChannel1	G1/0/1	DSW1 - G1/0/1		G0/1	DSW1 - G1/0/6
				G1/0/2	DSW1 - G1/0/2		G0/2	DSW2 - G1/0/6
				G1/0/7	CSW1 - G1/0/4	ASW5	F0/1	DNS Server
CSW2	G1/0/7	R1 - G0/1		G1/0/8	CSW2 - G1/0/4		G0/1	DSW3 - G1/0/3
PortChannel1	G1/0/1	CSW1 - G1/0/1	DSW3	G1/0/3	ASW5 - G0/1		G0/2	DSW4 - G1/0/3
	G1/0/2	CSW1 - G1/0/2		G1/0/4	ASW6 - G0/1	ASW6	F0/1	Email Server
	G1/0/3	DSW1 - G1/0/8		G1/0/5	ASW7 - G0/1		G0/1	DSW3 - G1/0/4
	G1/0/4	DSW2 - G1/0/8		G1/0/6	ASW8 - G0/1		G0/2	DSW4 - G1/0/4
	G1/0/5	DSW3 - G1/0/9		G1/0/7	ASW9 - G0/1	ASW7	F0/1	NTP Server
	G1/0/6	DSW4 - G1/0/9	PortChannel1	G1/0/1	DSW4 - G1/0/1		G0/1	DSW3 - G1/0/5
				G1/0/2	DSW4 - G1/0/2		G0/2	DSW4 - G1/0/5
				G1/0/8	CSW1 - G1/0/5	ASW8	F0/1	Syslog Server
				G1/0/9	CSW2 - G1/0/5		G0/1	DSW3 - G1/0/6
			DSW4	G1/0/3	ASW5 - G0/2		G0/2	DSW4 - G1/0/6
				G1/0/4	ASW6 - G0/2	ASW9	F0/1	AAA Server
				G1/0/5	ASW7 - G0/2		F0/2	LWAP2
				G1/0/6	ASW8 - G0/2		G0/1	DSW3 - G1/0/7
				G1/0/7	ASW9 - G0/2		G0/2	DSW4 - G1/0/7
			PortChannel1	G1/0/1	DSW3 - G1/0/1			
				G1/0/2	DSW3 - G1/0/2			
				G1/0/8	CSW1 - G1/0/6			
				G1/0/9	CSW2 - G1/0/6			

DEVICE	MODEL
CE-Router (x1)	Cisco 2911
Core Switches (x2)	Catalyst 3650 24 PS
Distribution Switches (x4)	Catalyst 3650 24 PS
Access Switches (x9)	Catalyst 2960

3.3 Logical Topology



3.4 VLAN Segmentation & IP Addressing Schema

Link	Subnet	Device/Interface	IP
CE-Router <> ISP-1	203.0.113.0/30	CE-Router - s0/3/0 ISP-1 - s0/3/0	203.0.113.2 203.0.113.1
CE-Router <> ISP-2	203.0.113.4/30	CE-Router - s0/3/1 ISP-2 - s0/3/0	203.0.113.6 203.0.113.5
CE-Router <> CSW1	10.0.0.32/30	CE-Router - g0/0 CSW1 - g1/0/7	10.0.0.33 10.0.0.34
CE-Router <> CSW2	10.0.0.36/30	CE-Router - g0/1 CSW2 - g1/0/7	10.0.0.37 10.0.0.38
CSW1 <> CSW2	10.0.0.40/30	CSW1 - Po1 CSW2 - Po1	10.0.0.41 10.0.0.42
CSW1 <> DSW1	10.0.0.44/30	CSW1 - g1/0/3 DSW1 - g1/0/7	10.0.0.45 10.0.0.46
CSW1 <> DSW2	10.0.0.48/30	CSW1 - g1/0/4 DSW2 - g1/0/7	10.0.0.49 10.0.0.50
CSW1 <> DSW3	10.0.0.52/30	CSW1 - g1/0/5 DSW3 - g1/0/8	10.0.0.53 10.0.0.54
CSW1 <> DSW4	10.0.0.56/30	CSW1 - g1/0/6 DSW4 - g1/0/8	10.0.0.57 10.0.0.58
CSW2 <> DSW1	10.0.0.60/30	CSW2 - g1/0/3 DSW1 - g1/0/8	10.0.0.61 10.0.0.62
CSW2 <> DSW2	10.0.0.64/30	CSW2 - g1/0/4 DSW2 - g1/0/8	10.0.0.65 10.0.0.66
CSW2 <> DSW3	10.0.0.68/30	CSW2 - g1/0/5 DSW3 - g1/0/9	10.0.0.69 10.0.0.70
CSW2 <> DSW4	10.0.0.72/30	CSW2 - g1/0/6 DSW4 - g1/0/9	10.0.0.73 10.0.0.74
Management (A) - VLAN 99	10.0.0.0/28	VIP (DSW1 <> DSW2) DSW1 (VLAN 99) DSW2 (VLAN 99) ASW1 (VLAN 99) ASW2 (VLAN 99) ASW3 (VLAN 99) ASW4 (VLAN 99) WLC1	10.0.0.1 10.0.0.2 10.0.0.3 10.0.0.4 10.0.0.5 10.0.0.6 10.0.0.7 10.0.0.8
Management (B) - VLAN 99	10.0.0.16/28	VIP (DSW3 <> DSW4) DSW3 (VLAN 99) DSW4 (VLAN 99) ASW5 (VLAN 99) ASW6 (VLAN 99) ASW7 (VLAN 99) ASW8 (VLAN 99) ASW9 (VLAN 99)	10.0.0.17 10.0.0.18 10.0.0.19 10.0.0.20 10.0.0.21 10.0.0.22 10.0.0.23 10.0.0.24

Link	Subnet	Device/Interface	IP
Loopbacks	10.0.0.76/32	CE-Router - L0	10.0.0.76
	10.0.0.77/32	CSW1 - L0	10.0.0.77
	10.0.0.78/32	CSW2 - L0	10.0.0.78
	10.0.0.79/32	DSW1 - L0	10.0.0.79
	10.0.0.80/32	DSW2 - L0	10.0.0.80
	10.0.0.81/32	DSW3 - L0	10.0.0.81
	10.0.0.82/32	DSW4 - L0	10.0.0.82
PCs (A) - VLAN 10	10.1.0.0/24	VIP	10.1.0.1
		DSW1 - VLAN 10	10.1.0.2
		DSW2 - VLAN 10	10.1.0.3
Phones (A) - VLAN 20	10.2.0.0/24	VIP	10.2.0.1
		DSW1 - VLAN 20	10.2.0.2
		DSW2 - VLAN 20	10.2.0.3
Printers (A) - VLAN 30	10.3.0.0/24	VIP	10.3.0.1
		DSW1 - VLAN 30	10.3.0.2
		DSW2 - VLAN 30	10.3.0.3
HR (A) - VLAN 40	10.4.0.0/24	VIP	10.4.0.1
		DSW1 - VLAN 40	10.4.0.2
		DSW2 - VLAN 40	10.4.0.3
DNS Server (B) - VLAN 50	10.5.0.0/24	VIP	10.5.0.1
		DSW3 - VLAN 50	10.5.0.2
		DSW4 - VLAN 50	10.5.0.3
		DNS Server	10.5.0.4
Email Server (B) - VLAN 60	10.6.0.0/24	VIP	10.6.0.1
		DSW3 - VLAN 60	10.6.0.2
		DSW4 - VLAN 60	10.6.0.3
		Email Server	10.6.0.4
NTP Server (B) - VLAN 70	10.7.0.0/24	VIP	10.7.0.1
		DSW3 - VLAN 70	10.7.0.2
		DSW4 - VLAN 70	10.7.0.3
		NTP Server	10.7.0.4
Syslog Server (B) - VLAN 80	10.8.0.0/24	VIP	10.8.0.1
		DSW3 - VLAN 80	10.8.0.2
		DSW4 - VLAN 80	10.8.0.3
		Syslog Server	10.8.0.4
AAA Server (B) - VLAN 90	10.9.0.0/24	VIP	10.9.0.1
		DSW3 - VLAN 90	10.9.0.2
		DSW4 - VLAN 90	10.9.0.3
		AAA Server	10.9.0.4
Wi-Fi (A) - VLAN 98	10.10.0.0/24	VIP	10.10.0.1
		DSW1 - VLAN 98	10.10.0.2
		DSW2 - VLAN 98	10.10.0.3
Wi-Fi (B) - VLAN 98	10.11.0.0/24	VIP	10.11.0.1
		DSW3 - VLAN 98	10.11.0.2
		DSW4 - VLAN 98	10.11.0.3

4.Design Details & Protocol Selection

. Campus Layering & Structural Architecture

- **Core Layer:** Dual Cisco Catalyst 3650 switches serving as the high-speed Layer 3 backbone. Operates as the central routing hub connecting inter-building traffic and the edge router.
- **Distribution Layer:** Catalyst 3650 switches handling Layer 2/3 boundary termination, HSRP default-gateway redundancy, and inter-VLAN routing policy enforcement.
- **Access Layer:** Catalyst 2960 switches providing port security, VLAN access, and Voice VLAN tagging (802.1Q) directly to end-hosts, printers, and Wireless Access Points (LWAPs).

. High Availability & Layer 2/3 Redundancy

- **HSRP (Hot Standby Router Protocol):**
 - Configured active/standby pairs across distribution switches (DSW1/DSW2 for HQ, DSW3/DSW4 for Production).
 - Virtual IPs assigned per VLAN (e.g., 10.1.0.1 for Sales, 10.0.0.1 for Management) to enable sub-second failover.
 - Preemption & Priority: Adjusted HSRP priority values to designate primary active gateways, ensuring predictable traffic flow.
- **LACP EtherChannel (802.3ad):**
 - Bundled dual Gigabit links into logical PortChannel1 trunks between Core switches (CSW1–CSW2) and Distribution switches (DSW1–DSW4).
 - Prevents Spanning Tree Protocol (STP) port blocking on critical interconnects while doubling available trunk bandwidth.

. Dynamic Routing Architecture (OSPF)

- **Single-Area OSPF (Area 0):** Enabled dynamic route propagation across Core switches and the Edge Router (CE-Router).
- **Point-to-Point Subnets:** Configured /30 links (10.0.0.32/30, 10.0.0.36/30, 10.0.0.40/30) to minimize route table bloat and reduce broadcast overhead.
- **Passive Interfaces:** Applied passive-interface commands on user-facing VLAN interfaces to prevent unauthorized OSPF neighbor adjacencies and reduce unnecessary routing traffic.

. Edge Connectivity, NAT/PAT & Security

- **PAT (Port Address Translation / NAT Overload):**
 - Mapped private internal subnets (10.0.0.0/8) to public WAN interfaces (203.0.113.0/30) on CE-Router.
 - Allowed multiple internal hosts to simultaneously access external resources through a single public IP address.
- **Access Control Lists (ACLs):**

- **Standard ACLs:** Defined transit rules identifying internal IP ranges permitted for NAT translation and SSH restriction.
- **Extended ACLs:** Restricted guest Wi-Fi traffic (VLAN 98) from traversing corporate subnets while permitting outbound HTTP/HTTPS access.

. Wireless & Management Infrastructure

- **WLC & LWAP Integration:** Deployed Lightweight APs (LWAP1, LWAP2) managed centrally via WLC1 to deliver seamless guest Wi-Fi coverage across both buildings.
- **Out-of-Band / Dedicated Management:** Isolated administrative management traffic onto dedicated /28 subnets (10.0.0.0/28 and 10.0.0.16/28) to protect switch management interfaces from standard user subnet traffic.

5.Configuration Documentation

5.1 Configuration Philosophy & Standards

The infrastructure configurations across all core, distribution, and access switches follow a strict baseline to ensure operational consistency, security, and maintainability:

- **Naming & Management Services:** Every device enforces SSH v2 (ip ssh version 2), localized admin accounts using modern password hashing (secret 5), domain mapping (yannickccna.com), and centralized management services including Syslog (10.8.0.4), NTP (10.7.0.4), and DNS (10.5.0.4).
- **In-Band Access Security:** VTY line access across all switches is locked down using Standard ACL 2 (access-class 2 in), restricting SSH access exclusively to authorized management subnets (10.0.0.0/28).
- **Unused Service Hardening:** Unnecessary broadcast and industrial protocols (such as PROFINET) are explicitly disabled across the switching domain to reduce system overhead and attack vectors.

5.2 Core Switching Layer (CSW1 / CSW2)

The Core Layer provides high-speed, non-blocking Layer 3 packet switching across the enterprise backbone using point-to-point links and dual-stack OSPF routing.

Cisco CLI

```
! =====  
! LAYER 3 ETHERCHANNEL & DUAL-STACK OSPF (CSW1 HIGHLIGHT)  
! =====  
! Routed Inter-Switch Core Bundle  
  
interface Port-channel1  
  
description Routed Core Inter-Switch Link  
  
no switchport  
  
ip address 10.0.0.41 255.255.255.252  
  
ipv6 address 2001:DB8:E1::/126  
  
ipv6 ospf 1 area 0  
  
  
interface GigabitEthernet1/0/1
```

description Core Backbone Link 1

no switchport

channel-protocol lacp

channel-group 1 mode active

! Downlink Routed Point-to-Point Interface to DSW1

interface GigabitEthernet1/0/3

description Link to DSW1

no switchport

ip address 10.0.0.45 255.255.255.252

ip ospf network point-to-point

ipv6 address 2001:DB8:E2::/126

ipv6 ospf 1 area 0

! Area 0 IPv4/IPv6 OSPF Routing

router ospf 1

router-id 10.0.0.77

passive-interface Loopback0

network 10.0.0.77 0.0.0.0 area 0

network 10.0.0.32 0.0.0.3 area 0 ! WAN Edge Uplink

network 10.0.0.40 0.0.0.3 area 0 ! Core Inter-Switch Link

network 10.0.0.44 0.0.0.3 area 0 ! DSW1 Interconnect

ipv6 router ospf 1

router-id 10.0.0.77

Key Core Architecture Highlights

1. **Routed LACP EtherChannels:** The core inter-switch link (Port-channel1) operates at Layer 3 (no switchport), combining multiple physical links with LACP while maintaining discrete /30 IPv4 and /126 IPv6 subnets to prevent L2 switching loops at the core.
2. **Point-to-Point OSPF Adjacencies:** Inter-switch links explicitly enforce ip ospf network point-to-point to eliminate DR/BDR election overhead and accelerate SPF convergence.

5.3 Distribution Switching Layer (DSW1 / DSW2)

The Distribution Layer serves as the Layer 2/Layer 3 boundary, managing First-Hop Redundancy Protocols (FHRP), DHCP relay services, and STP root bridge placement.

Cisco CLI

! =====

! HSRP V2 ACTIVE GATEWAY & OSPF PASSIVE SVIs (DSW1 HIGHLIGHT)

! =====

! Spanning-Tree Root Bridge Priority

spanning-tree mode rapid-pvst

spanning-tree vlan 10,20,30 priority 0

spanning-tree vlan 40,98-99 priority 4096

! Active HSRP Gateway & DHCP Helper

interface Vlan10

description Sales Subnet Gateway (Primary)

ip address 10.1.0.2 255.255.255.0

ip helper-address 10.0.0.33

standby version 2

standby 1 ip 10.1.0.1

standby 1 priority 105

standby 1 preempt

! Passive OSPF SVI Advertisements

```

router ospf 1

router-id 10.0.0.79

passive-interface Loopback0

passive-interface Vlan10

passive-interface Vlan20

passive-interface Vlan30

passive-interface Vlan40

passive-interface Vlan98

network 10.0.0.79 0.0.0.0 area 0

network 10.0.0.44 0.0.0.3 area 0  ! Core Uplink CSW1

network 10.0.0.60 0.0.0.3 area 0  ! Core Uplink CSW2

network 10.1.0.0 0.0.0.255 area 0  ! Sales Subnet /24

```

Key Distribution Architecture Highlights

3. **Active/Standby Gateway Load Sharing:** DSW1 acts as the active HSRP gateway (priority 105, preempt) and Rapid-PVST+ Primary Root Bridge (priority 0) for user VLANs 10, 20, and 30, while serving as secondary/standby for VLANs 40, 98, and 99.
4. **Centralized DHCP Relay:** Each user SVI includes ip helper-address 10.0.0.33 to forward DHCP broadcast traffic across Layer 3 boundaries to the core DHCP server (CE-Router).
5. **OSPF Control Plane Security:** User access SVIs are explicitly defined as passive-interface within OSPF process 1. This advertises the host subnets to the core without broadcasting OSPF hellos down to access switches.

5.4 Access Switching Layer (ASW1–ASW4)

The Access Layer enforces edge security policies, VLAN segmentation, and port protection for end devices.

Cisco CLI

! =====

! LAYER 2 DEFENSE-IN-DEPTH SECURITY (ASW1 HIGHLIGHT)

! =====

! Edge Security Services

```
ip arp inspection vlan 10,20,30,40,98,99
ip arp inspection validate src-mac dst-mac ip
ip dhcp snooping vlan 10,20,30,40,98,99
no ip dhcp snooping information option
ip dhcp snooping
```

! Edge Access Port with Voice VLAN & Sticky Port Security

```
interface FastEthernet0/1
description Workstation + IP Phone (Sales)
switchport access vlan 10
switchport voice vlan 20
switchport mode access
switchport nonegotiate
switchport port-security
switchport port-security maximum 3
switchport port-security violation restrict
switchport port-security mac-address sticky
spanning-tree portfast
spanning-tree bpduguard enable
```

! Hardened Trunk Uplink to Distribution

```
interface GigabitEthernet0/1
description Uplink to DSW1
switchport mode trunk
switchport trunk native vlan 999
switchport trunk allowed vlan 10,20,30,40,98-99
switchport nonegotiate
```

ip dhcp snooping trust

ip arp inspection trust

Key Access Architecture Highlights

6. **Layer 2 Defense-in-Depth: DHCP Snooping** prevents rogue DHCP server injection, while **Dynamic ARP Inspection (DAI)** validates ARP packets against the snooping binding database to block ARP poisoning/man-in-the-middle attacks.
7. **Port Security & STP Safeguards:** User ports restrict MAC addresses (maximum 3, violation restrict) using sticky learning. spanning-tree portfast accelerates access port transition times, paired with bpduguard enable to shut down ports if unauthorized switches are connected.
8. **VLAN Hopping Mitigation:** Trunk links explicitly restrict allowed VLANs (10,20,30,40,98-99) and isolate untagged frame processing using an unused native VLAN (native vlan 999).

5.5 Device Management & IP Summary Table

Device	Role	Management IP	Routing Protocol / ID	Key Configured Roles
CSW1	Core Switch	Loopback 10.0.0.77/32	OSPFv2 / OSPFv3 (10.0.0.77)	L3 Core Interconnect, L3 EtherChannel (Po1)
CSW2	Core Switch	Loopback 10.0.0.78/32	OSPFv2 / OSPFv3 (10.0.0.78)	L3 Core Interconnect, L3 EtherChannel (Po1)
DSW1	Distribution	10.0.0.2/28 (VLAN 99)	OSPFv2 / OSPFv3 (10.0.0.79)	Active HSRP (VLANs 10-30), Primary STP Root, DHCP Relay
DSW2	Distribution	10.0.0.3/28 (VLAN 99)	OSPFv2 / OSPFv3 (10.0.0.80)	Active HSRP (VLANs 40, 98-99), Secondary STP Root, DHCP Relay
ASW1	Access Switch	10.0.0.4/28 (VLAN 99)	Static Default Gateway (10.0.0.1)	DAI, DHCP Snooping, Port Security, Voice VLAN, BPDU Guard

6. Testing & Validation

6.1 Test Strategy & Execution Methodology

To verify that the multi-tier enterprise network meets all operational and business requirements, a structured test plan was executed across four primary domain areas:

1. **Layer 2 Connectivity & Security:** Validating VLAN isolation, Trunking, Rapid-PVST+ root placement, DHCP Snooping, Dynamic ARP Inspection (DAI), and Port Security.
2. **First-Hop Redundancy Protocols (FHRP):** Verifying HSRP v2 state transitions, virtual IP gateways, active/standby failover times, and preemption.
3. **Core & Distribution Dynamic Routing:** Confirming dual-stack OSPF area 0 neighbor adjacencies, path cost metrics, passive interfaces, and route table convergence.
4. **End-to-End Traffic & Services:** Testing DHCP IP allocation, DNS resolution, Syslog logging, NTP time synchronization, and administrative SSH access control.

6.2 Test Plan & Execution Results

Test Case 1: Layer 2 Loop Prevention & Root Bridge Placement

- **Objective:** Ensure Rapid-PVST+ prevents switching loops and enforces designated Root Bridge primary placement.
- **Verification Commands:** ASW1# show spanning-tree vlan 10 ; DSW1# show spanning-tree vlan 10
- **Expected Result:** DSW1 reports as the Root Bridge for VLAN 10 (Priority 0). ASW1 identifies Gi0/1 (uplink to DSW1) as its Root Port (Forwarding) and Gi0/2 as Alternate Port (Blocking).
- **Status: PASS**

Test Case 2: Port Security & Rogue Device Mitigation

- **Objective:** Verify that unauthorized devices attempting to connect to access ports are restricted and logged.
- **Verification Procedure:**
 1. Connect a device with an unapproved MAC address to FastEthernet0/1 on ASW1.
 2. Execute verification command: ASW1# show port-security interface fastEthernet 0/1
- **Expected Result:** Port security violation counter increments. The switch restricts traffic from the unauthorized MAC address while keeping the port operational for allowed sticky MAC addresses. syslog outputs a security restriction alert to 10.8.0.4.
- **Status: PASS**

Test Case 3: First-Hop Redundancy (HSRP v2) & Gateway Preemption

- **Objective:** Validate active/standby gateway load sharing and immediate failover during uplink failure.
- **Verification Procedure:**

1. Check HSRP states across distribution switches: DSW1# show standby brief ; DSW2# show standby brief
 2. Initiate a simulated link failover by shutting down DSW1's SVI or core uplink: DSW1(config)# interface vlan 10 ; DSW1(config-if)# shutdown
 3. Perform a continuous ping from a Sales host (10.1.0.50) to the core loopback (10.0.0.77).
- **Expected Result:** DSW1 initially reports Active for VLANs 10–30 and Standby for VLAN 40. Upon interface shutdown, DSW2 transitions from Standby to Active in under 3 seconds with minimal packet loss. Re-enabling the interface triggers DSW1 preemption back to Active.
 - **Status: PASS**

Test Case 4: OSPFv2 / OSPFv3 Dual-Stack Routing Convergence

- **Objective:** Ensure OSPF establishes full point-to-point adjacencies across the core and correctly advertises IPv4/IPv6 subnets.
- **Verification Commands:** CSW1# show ip ospf neighbor ; CSW1# show ipv6 ospf neighbor ; CSW1# show ip route ospf
- **Expected Result:** CSW1 displays FULL/ - neighbor states across all point-to-point interface links (Gi1/0/3, Gi1/0/7, Port-channel1). The routing table populates all remote VLAN subnets (10.1.0.0/24, 10.2.0.0/24, etc.) with Loopback0 interfaces set as passive.
- **Status: PASS**

Test Case 5: Infrastructure Services & In-Band Management ACL

- **Objective:** Verify DHCP relay, Syslog forwarding, and SSH management access restriction.
- **Verification Commands:** ASW1# show ip dhcp snooping binding ; DSW1#show running-config | include ip helper ; ASW1# ssh -l yannick 10.0.0.2 (secret: cisco)
- **Expected Result:**
 - Workstations dynamically lease IP addresses from 10.0.0.33 via SVI ip helper-address.
 - SSH connections originating from the management subnet (10.0.0.0/27) succeed.
 - SSH connections originating from unauthorized user subnets (e.g., VLAN 10 10.1.0.0/24) are blocked by Access-List 2 (Connection refused by remote host).
- **Status: PASS**

6.3 Test & Validation Summary Table

Test ID	Test Category	Target Device(s)	Success Criteria	Result
TC-01	Spanning Tree	ASW1, DSW1, DSW2	Root bridge assigned to DSW1 (VLAN 10-30); loops blocked on redundant trunks.	PASS
TC-02	Layer 2 Security	ASW1	DAI & DHCP Snooping drop spoofed ARP/DHCP packets; Port Security restricts extra MACs.	PASS
TC-03	Gateway Failover	DSW1, DSW2	HSRP failover to DSW2 occurs in < 3 seconds upon link failure; preemption restored on recovery.	PASS
TC-04	Dual-Stack OSPF	CSW1, CSW2, DSW1-2	OSPFv2/v3 adjacencies reach FULL state over point-to-point links; subnets properly routed.	PASS
TC-05	Service Relay	DSW1, DSW2	DHCP discover packets successfully relayed to 10.0.0.33; clients acquire correct leases.	PASS
TC-06	Management Security	All Devices	VTY ACL blocks unauthorized SSH attempts outside 10.0.0.0/27. Syslog messages forwarded to 10.8.0.4.	PASS

7.Key Takeaways & Lessons Learned

Executing this multi-tier enterprise network design provided hands-on experience in balancing hierarchical network architecture, high availability, control plane efficiency, and defense-in-depth security. Below are the primary technical insights and engineering principles reinforced throughout this project.

7.1 Layer 3 Core & Dynamic Routing Optimization

- **Routed EtherChannels vs. Layer 2 Bundles:** Transitioning the core inter-switch link to a Layer 3 routed EtherChannel eliminated Spanning Tree dependency at the backbone layer. This design choice prevents unexpected convergence delays and optimizes link bandwidth using Layer 3 equal-cost multi-pathing (ECMP).
- **Point-to-Point OSPF Network Types:** Explicitly configuring `ip ospf network point-to-point` on point-to-point links bypassed Designated Router (DR) and Backup Designated Router (BDR) elections. This reduced control plane overhead and accelerated SPF calculation times during link state changes.
- **OSPF Wildcard Precision & Control:** Defining exact wildcard masks (0.0.0.255 for /24 subnets) and marking user-facing SVIs as passive-interface prevented rogue OSPF adjacency formation while keeping routing tables clean and secure.

7.2 High Availability & Traffic Alignment

- **Symmetric Spanning Tree & HSRP Alignment:** A critical takeaway was aligning First-Hop Redundancy Protocol (HSRP) priority with Rapid-PVST+ Root Bridge placement. Configuring DSW1 as the active HSRP gateway (Priority 105) and Primary Root Bridge (Priority 0) for VLANs 10, 20, and 30 ensures that Layer 2 traffic flows natively toward the active Layer 3 gateway without traversing inter-switch trunk links unnecessarily.
- **FHRP Preemption:** Enabling `standby preempt` ensures that primary gateways automatically reclaim active forwarding status upon recovering from an outage, maintaining deterministic traffic paths.

7.3 Edge Security & Defense-in-Depth

- **Mitigating Layer 2 Attack Vectors:** Implementing **DHCP Snooping** alongside **Dynamic ARP Inspection (DAI)** proved essential for securing access ports against DHCP spoofing, rogue servers, and ARP poisoning attacks.
- **VLAN Hopping & Native VLAN Security:** Standardizing native VLAN isolation across all trunks (using an unused ID such as native vlan 999) highlighted the necessity of consistent trunk definitions across access and distribution switches to prevent syslog mismatches and frame double-tagging exploits.
- **In-Band Management Isolation:** Restricting VTY line access via Standard ACLs and locking down administrative management exclusively to SSH v2 provided crucial infrastructure hardening against unauthenticated edge access attempts.

8. Conclusion & Future Enhancements

8.1 Executive Conclusion

This project successfully designed, implemented, and validated a robust, 3-tier campus network architecture tailored for enterprise operations. The design delivers predictable scalability, high availability, and comprehensive defense-in-depth security. Key achievements of this deployment include:

- **Resilient Infrastructure:** Rapid failover achieved through synchronized HSRP v2 active/standby gateways and Rapid-PVST+ root bridge alignment.
- **Optimized Routing Core:** High-speed, loop-free packet transport across a dual-stack OSPF core using Layer 3 routed EtherChannels and point-to-point subnets.
- **Hardened Edge Defense:** Comprehensive Layer 2 threat mitigation enforcing DHCP Snooping, Dynamic ARP Inspection (DAI), sticky Port Security, and explicit native VLAN isolation.
- **Deterministic Administration:** Centralized management via SSH v2, VTY access restriction ACLs, NTP synchronization, and Syslog auditing.

The empirical testing and validation results confirm that the network infrastructure meets all initial performance, security, and redundancy requirements, providing a resilient foundation for mission-critical business applications.

8.2 Future Enhancements

While the current design provides a solid baseline for campus operations, enterprise networks must continuously evolve to accommodate growing scale, modern security paradigms, and automated operations. Recommended future initiatives include:

1. **Network Automation & Infrastructure-as-Code (IaC):**
 - Adopt Python (Netmiko) or Ansible playbooks for automated device configuration, compliance auditing, and OS patch management.
 - Store device configurations in a Git repository connected to a CI/CD pipeline for automated testing in a staging environment prior to production deployment.
2. **Software-Defined Access (SD-Access) & VXLAN Overlay:**
 - Modernize the Layer 2/Layer 3 boundary by migrating toward a VXLAN-EVPN fabric overlay.
 - Implement micro-segmentation to enforce zero-trust access control across users and IoT devices regardless of physical location.
3. **Enhanced Network Visibility & Telemetry:**
 - Upgrade from legacy SNMP/Syslog monitoring to Streaming Telemetry (NETCONF/RESTCONF) for real-time performance analytics.
 - Deploy NetFlow/IPFIX collectors to perform deep traffic analysis and detect anomalous lateral movement within the network.